

Capture the Flag

Your Student Portal is now dynamic thanks to interpolation.

But there's a problem.

Your announcement component currently has something like:

```
announcementTitle1 = "Enrollment is Now Open";  
announcementBody1 = "Enrollment for the next semester is now  
available.";
```

```
announcementTitle2 = "Intramurals 2026";  
announcementBody2 = "Registration is now open.";
```

```
announcementTitle3 = "Library Schedule";  
announcementBody3 = "The library will be open until 8 PM.";
```

And your HTML has to repeat the same card structure three times.

The school now has **20 announcements**.

Your mission:

Stop repeating yourself. Make Angular generate the cards for you.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/ngFor`



Capture

- Show your current Git branch.

```
woj@2 my-app % git checkout -b feature/ngFor  
Switched to a new branch 'feature/ngFor'  
woj@2 my-app % █
```

FLAG 1 — The Repetition Problem

Open your announcement component.

Look at your existing announcement cards.

You should currently have something similar to:

```
<div class="announcement-card">
  <h3>{{ announcementTitle1 }}</h3>
  <p>{{ announcementBody1 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle2 }}</h3>
  <p>{{ announcementBody2 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle3 }}</h3>
  <p>{{ announcementBody3 }}</p>
</div>
```

Mission

Imagine the school gives you **10 more announcements**.

Would you want to copy and paste the card **10 more times**?

Don't change anything yet.



Capture: Your repeated announcement cards.

```
src > app > components > announcement-card > announcement-card.html > div.announcement
1  <div class="announcement-card">
2    <h3 class="card-title">Welcome {{AnnouncementTitle1}}</h3>
3    <p class="card-description">{{AnnouncementBody1}}</p>
4    <span class="card-date">August 4, 2026</span>
5  </div>
6
7  <div class="announcement-card">
8    <h3 class="card-title">Welcome {{AnnouncementTitle2}}</h3>
9    <p class="card-description">{{AnnouncementBody2}}</p>
10   <span class="card-date">August 4, 2026</span>
11 </div>
12
13 <div class="announcement-card">
14   <h3 class="card-title">Welcome {{AnnouncementTitle3}}</h3>
15   <p class="card-description">{{AnnouncementBody3}}</p>
16   <span class="card-date">August 4, 2026</span>
17 </div>
18 <div class="announcement-card">
19   <h3 class="card-title">Welcome {{AnnouncementTitle4}}</h3>
20   <p class="card-description">{{AnnouncementBody4}}</p>
21   <span class="card-date">August 4, 2026</span>
22 </div>
23
24 <div class="announcement-card">
25   <h3 class="card-title">Welcome {{AnnouncementTitle5}}</h3>
26   <p class="card-description">{{AnnouncementBody5}}</p>
27   <span class="card-date">August 4, 2026</span>
28 </div>
29
```

FLAG 2 — Put the Data Together

Instead of creating separate variables, create **one array**.

Research Hint

An array can contain multiple objects.

You don't need to memorize the syntax.

Search:

Angular array of objects



Capture: Your `announcements` array.

```
announcements = [  
  {  
    title: "Enrollment is now open",  
    body: "Enroll now"  
  },  
  {  
    title: "Intramurals 2026",  
    body: "Join us"  
  },  
  {  
    title: "Library is now open",  
    body: "8:00 AM - 5:00 PM"  
  },  
]
```

FLAG 3 — Meet **ngFor*

Now comes the important part.

You have:

```
announcements = [  
  { ... },  
  { ... },  
  { ... }  
];
```

How can Angular create one card **for every item** in that array?

Hint

Research:

**Angular ngFor*

You should discover syntax similar to:

```
*ngFor="let announcement of announcements"
```

Your mission is to use it on your announcement card.

Instead of having:

```
<div class="announcement-card">
```

repeated three times, you should have **one card template** that Angular repeats.



Capture:

- Your updated code.

```
<section class="announcements-section">
  @for(announcement of announcements; track announcement.title){
    <app-announcement-card [announcement] = "announcement"></app-announcement-card>
  }
</section>
```

FLAG 4 — Display the Current Announcement

You now have a loop variable.

Inside the repeated card, display title and body:

Your browser should now display all eight announcements.

Important

You should have:

1 array

1 card template

8 displayed cards

Not:

8 arrays

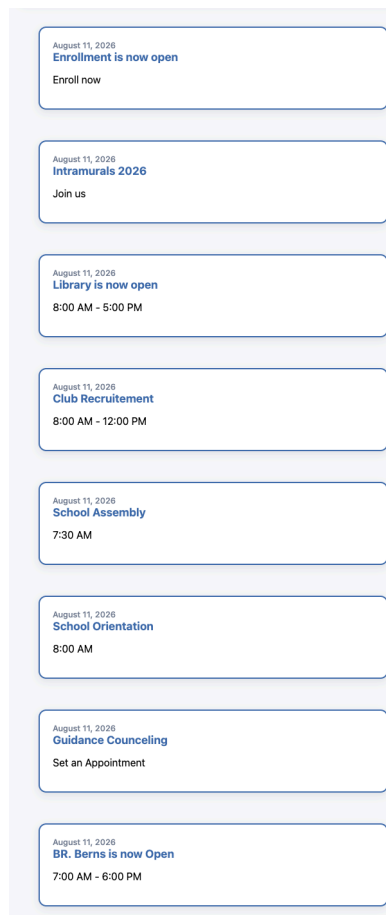
8 card templates

 Capture:

- Browser showing all announcements.
- Updated code.

```
announcements = [
  {
    title: "Enrollment is now open",
    body: "Enroll now"
  },
  {
    title: "Intramurals 2026",
    body: "Join us"
  },
  {
    title: "Library is now open",
    body: "8:00 AM - 5:00 PM"
  },
  {
    title: "Club Recrutement",
    body: "8:00 AM - 12:00 PM"
  },
  {
    title: "School Assembly",
    body: "7:30 AM"
  },
  {
    title: "School Orientation",
    body: "8:00 AM"
  },
  {
    title: "Guidance Counseling ",
    body: "Set an Appointment"
  },
  {
    title: "BR. Berns is now Open",
    body: "7:00 AM - 6:00 PM"
  },
]
```

```
<div class="card">
  <div class="card-header">
    <span class="card-date">{{ todayDate | date:'MMMM d, y' }}</span>
  </div>
  <div class="card-body">
    <h3 class="card-title">{{ announcement?.title }}</h3>
    <p class="card-text">{{ announcement?.body }}</p>
  </div>
</div>
```



FLAG 5 — Add More Data

Now test whether your loop actually works.

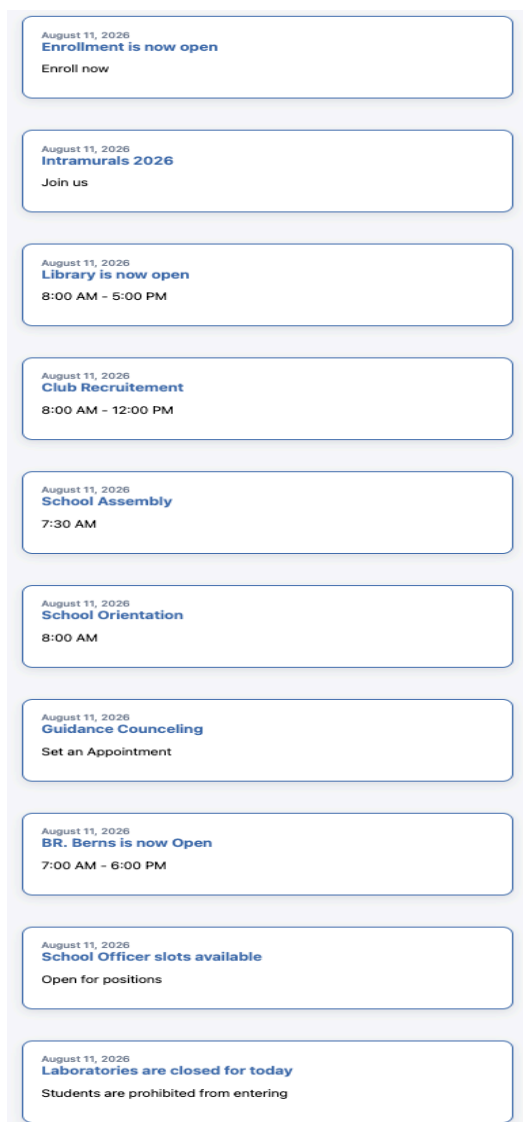
Add **two more announcements** to the array.

You should **not change your HTML**.

If two additional cards automatically appear:

 **FLAG CAPTURED**

 **Capture:** Browser showing all ten announcements.



FLAG 6 — Make the Cards More Realistic

Your announcement objects currently contain only:

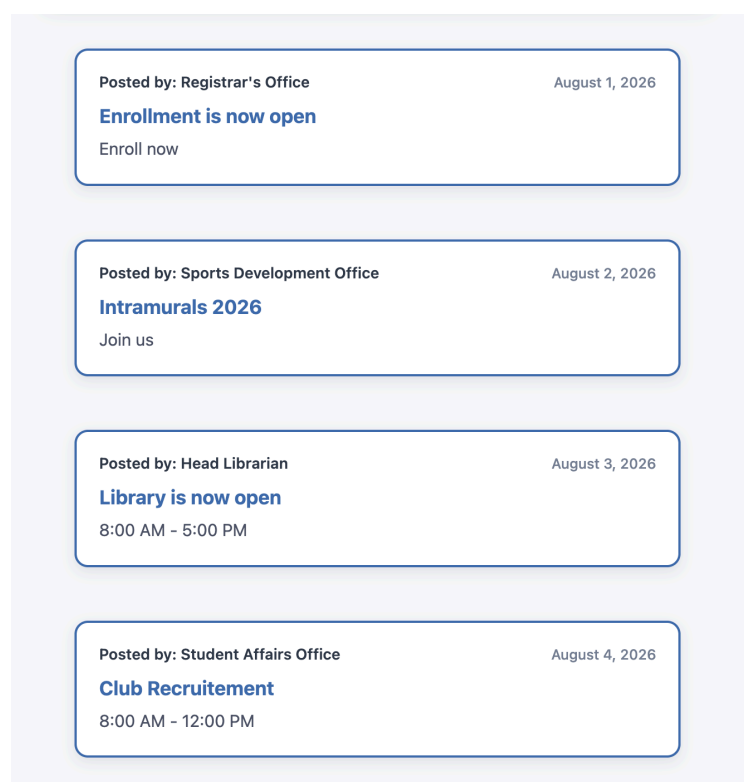
title
body

Add more information: author and date.

Update your card to display all fields.

Use **interpolation** for all values.

 **Capture:** Your richer announcement cards.



Posted by: Registrar's Office August 1, 2026

Enrollment is now open

Enroll now

Posted by: Sports Development Office August 2, 2026

Intramurals 2026

Join us

Posted by: Head Librarian August 3, 2026

Library is now open

8:00 AM - 5:00 PM

Posted by: Student Affairs Office August 4, 2026

Club Recrutement

8:00 AM - 12:00 PM

FLAG 7 — Student List

Announcements aren't the only thing that can be repeated.

Create another array named students.

Create a simple student card.

Use `*ngFor` to display every student.

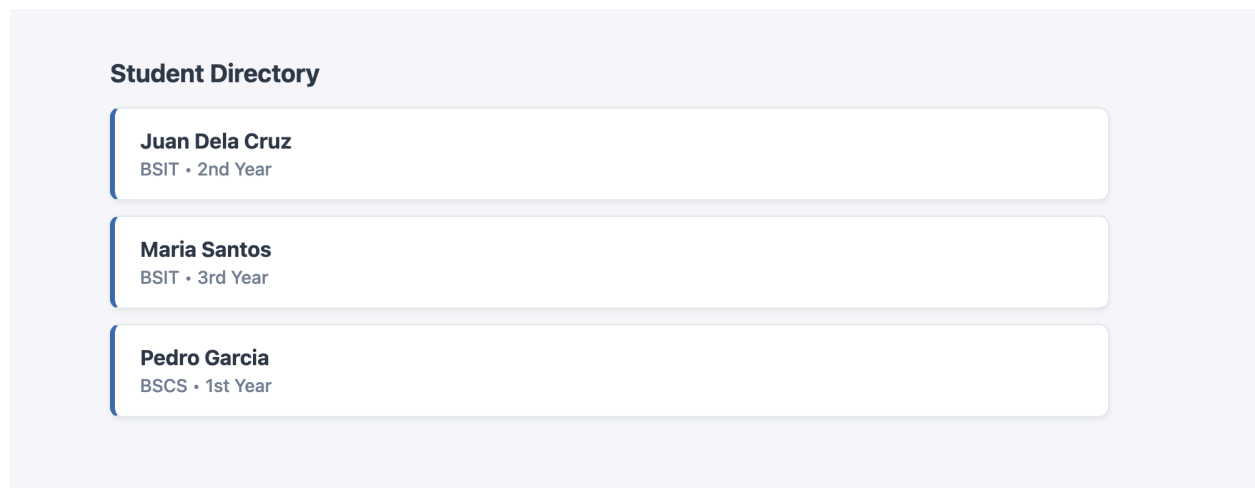
Expected example:

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

 **Capture:** Student list.



FLAG 8 — One Template, Many Students

Add another student to your array.

Do **not** modify the HTML.

The new student should automatically appear.

This is the important concept:

DATA CHANGES



ARRAY CHANGES



ANGULAR GENERATES THE UI



Capture: Before and after adding the student

Student Directory

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

Student Directory

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

Ana Reyes
BSIS • 4th Year

FLAG 9 — Research Challenge

I will **not** provide the next syntax.

Research:

**Angular ngFor index*

Find out how to access the position of the current item.

Use it to display:

#1 Juan Dela Cruz

#2 Maria Santos

#3 Pedro Garcia

Without hardcoding the numbers.

Hint:

let i = index



Capture:

- Your numbered student list.
- Your code

```
1 <app-header></app-header>
2
3 <main class="dashboard-body">
4   <app-profile-card></app-profile-card>
5
6   <section class="announcements-section">
7     @for (announcement of announcements; track announcement.title) {
8       <app-announcement-card [announcement]="announcement"></app-announcement-card>
9     }
10  </section>
11
12  <section class="students-section">
13    <h2>Student List</h2>
14    <div class="student-card" *ngFor="let student of students; let i = index">
15      <h3 class="student-name">#{{ i + 1 }} {{ student.name }}</h3>
16      <p class="student-info">{{ student.course }} • {{ student.year }}</p>
17    </div>
18  </section>
19 </main>
20
21 <app-footer></app-footer>
22
23 <router-outlet />
```

Student List

#1 Juan Dela Cruz

BSIT • 2nd Year

#2 Maria Santos

BSIT • 3rd Year

#3 Pedro Garcia

BSCS • 1st Year

#4 Ana Reyes

BSIS • 4th Year

FLAG 10 — Empty List

Now test what happens when there are no announcements.

Change:

```
announcements = [];
```

What happens?

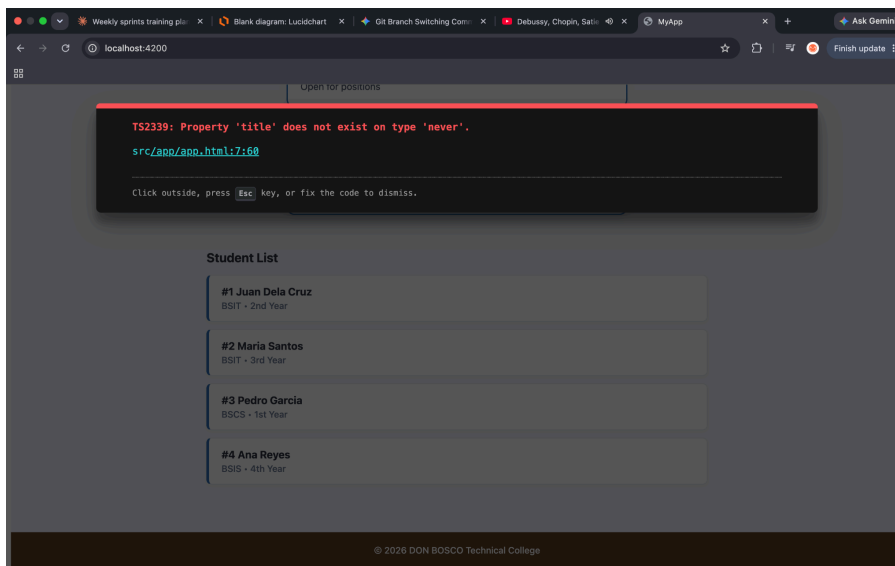
Write a short observation in your documentation:

What happens when the array is empty?

Answer: When removing the values inside the array, an error will appear. Because when the array is empty, Angular has no data to loop through, so it renders nothing on the screen.

Then restore your announcements.

 **Capture:** Your experiment and observation.



FLAG 11 — Dynamic Student Portal

Your Student Portal should now contain:

 **Your Student Profile**

Still using interpolation:

Name

Student ID
Course
Year
Grade
Academic Standing
Email

Announcements

Generated from:

announcements = [...]

Using:

*ngFor

Student List

Generated from:

students = [...]

Using:

*ngFor and index

Your final requirement

Final Screenshot

- Entire browser UI of all requirements in this activity

Don Bosco Student Portal



Authorize Signature

MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle S. Rizal

ID No : 23017245
Course : BSIT - IV
Date : July 29, 2026
Section : Undefined
Nickname : KURT
Email : kurtwojtyle.rizal@dbtc-cebu.edu.ph
Grade : Pass



Posted by: Registrar's Office August 1, 2026
Enrollment is now open
Enroll now

Posted by: Sports Development Office August 2, 2026
Intramurals 2026
Join us

Posted by: Academic Affairs August 6, 2026
School Orientation
8:00 AM

Student List

#1 Juan Dela Cruz
BSIT - 2nd Year

#2 Maria Santos
BSIT - 3rd Year

#3 Pedro Garcia
BSCS - 1st Year

#4 Ana Reyes
BSIS - 4th Year

FLAG 12 — Commit & Push

Final Screenshot

- Successful push to feature/ngFor

```
woj@2 my-app % git commit -m "Act 4: Loops"
[feature/ngFor 263cf7d] Act 4: Loops
 6 files changed, 127 insertions(+), 104 deletions(-)
woj@2 my-app % git push origin feature/ngFor
Enumerating objects: 28, done.
Counting objects: 100% (28/28), done.
Delta compression using up to 8 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (16/16), 79.98 KiB | 26.66 MiB/s, done.
Total 16 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 4 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL.git
remote:
remote: Create a pull request for 'feature/ngFor' on GitHub by visiting:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL/pull/new/feature/ngFor
remote:
To https://github.com/W0jkurt/ITE403A_RIZAL.git
 * [new branch]      feature/ngFor -> feature/ngFor
woj@2 my-app %
```